Example: Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation given by

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5x+6y \\ -3x+4y \end{bmatrix}$$

$\vec{e}_1 = -\vec{b}_1 + \vec{b}_2$

$\vec{e}_2 = 2\vec{b}_1 - \vec{b}_2$

and let $B = \{\vec{e}_1, \vec{e}_2\} = \left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}$ and $B' = \{\vec{b}_1, \vec{b}_2\} = \left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}$ be the bases

(I) Compute matrix A_1 of T with respect to B.

$$\left.\begin{aligned} T\begin{bmatrix} 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} -5 \\ -3 \end{bmatrix} \\ T\begin{bmatrix} 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 6 \\ 4 \end{bmatrix} \end{aligned}\right\} \Longrightarrow A_1 = \begin{bmatrix} -5 & 6 \\ -3 & 4 \end{bmatrix}$$

(II) Compute matrix A_2 of T with respect to B'.

$$\left.\begin{aligned} T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad &\text{and} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}_{B'} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ T\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \end{bmatrix} \quad &\text{and} \quad \begin{bmatrix} -4 \\ -2 \end{bmatrix}_{B'} = \begin{bmatrix} 0 \\ -2 \end{bmatrix} \end{aligned}\right\} \Longrightarrow A_2 = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$$

Example: Let $T:\mathbb{R}^2 \longrightarrow \mathbb{R}^2$ be the linear Transformation given by

$$P_{B'\leftarrow B} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \qquad T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5x+6y \\ -3x+4y \end{bmatrix} \qquad \boxed{\begin{aligned} \vec{e}_1 &= -\vec{b}_1 + \vec{b}_2 \\ \vec{e}_2 &= 2\vec{b}_1 - \vec{b}_2 \end{aligned}}$$

and let $B = \{\vec{e}_1, \vec{e}_2\} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $B' = \{\vec{b}_1, \vec{b}_2\} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ be the bases

(I) Compute matrix A_1 of T with respect to B.

$$\left. \begin{aligned} T\begin{bmatrix} 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} -5 \\ -3 \end{bmatrix} \\ T\begin{bmatrix} 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 6 \\ 4 \end{bmatrix} \end{aligned} \right\} \Longrightarrow A_1 = \begin{bmatrix} -5 & 6 \\ -3 & 4 \end{bmatrix} \qquad A_1 [\vec{u}]_B = [\vec{v}]_B$$

(II) Compute matrix A_2 of T with respect to B'.

$$P_{B'\leftarrow B} A_1 \underbrace{P^{-1}_{B'\leftarrow B} P_{B'\leftarrow B}}_{I} [\vec{u}]_B = P_{B'\leftarrow B} [\vec{v}]_B \qquad \boxed{A_2 = P_{B'\leftarrow B} A_1 P^{-1}_{B'\leftarrow B}}$$

$$P_{B'\leftarrow B} A_1 P^{-1}_{B'\leftarrow B} \, P_{B'\leftarrow B} [\vec{u}]_B = P_{B'\leftarrow B} [\vec{v}]_B \Longrightarrow A_2 [\vec{u}]_{B'} = [\vec{v}]_{B'}$$

$$T[\vec{x}]_{B'} = A'[\vec{x}]_{B'}$$

$$T(x\vec{b}_1 + y\vec{b}_2) = x\,T(\vec{b}_1) + y\,T(\vec{b}_2) = x\vec{b}_1 - 2y\,\vec{b}_2$$

(III) Find the matrix A_3 of T if the basis B and B' are used for domain and codomain respectively.

First Method

$$[T(\vec{e}_1)]_{B'} = \begin{bmatrix}-5\\-3\end{bmatrix}_{B'} = -1\begin{bmatrix}1\\1\end{bmatrix} - 2\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}-1\\-2\end{bmatrix}$$

$$[T(\vec{e}_2)]_{B'} = \begin{bmatrix}6\\4\end{bmatrix}_{B'} = 2\begin{bmatrix}1\\1\end{bmatrix} + 2\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}2\\2\end{bmatrix}$$

$$A_3 = \begin{bmatrix}-1 & 2\\-2 & 2\end{bmatrix}$$

$$\begin{bmatrix}-1 & 2\\-2 & 2\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}_B = \begin{bmatrix}-x+2y\\-2x+2y\end{bmatrix}_{B'}$$

$$T\begin{bmatrix}x\\y\end{bmatrix}_B = \begin{bmatrix}-x+2y\\-2x+2y\end{bmatrix}_{B'}$$

Second Method

$$A_1[\vec{u}]_B = [\vec{v}]_B \qquad P_{B'\leftarrow B}A_1[\vec{u}]_B = P_{B'\leftarrow B}[\vec{v}]_B$$

$$P_{B'\leftarrow B}A_1[\vec{u}]_B = [v]_{B'}$$

$$\boxed{A_3 = P_{B'\leftarrow B}A_1} = \begin{bmatrix}-1 & 2\\1 & -1\end{bmatrix}\begin{bmatrix}-5 & 6\\-3 & 4\end{bmatrix} = \begin{bmatrix}-1 & 2\\-2 & 2\end{bmatrix}$$

Third Method

$$\boxed{\begin{aligned}\vec{e}_1 &= -\vec{b}_1+\vec{b}_2\\ \vec{e}_2 &= 2\vec{b}_1-\vec{b}_2\end{aligned}}$$

$$\begin{aligned}
T(x\vec{e}_1+y\vec{e}_2) &= xT(\vec{e}_1)+yT(\vec{e}_2)\\
&= x(-5\vec{e}_1-5\vec{e}_2)+y(6\vec{e}_1+4\vec{e}_2)\\
&= x(5\vec{b}_1-5\vec{b}_2-6\vec{b}_1+3\vec{b}_2)+y(-6\vec{b}_1+6\vec{b}_2+8\vec{b}_1-4\vec{b}_2)\\
&= x(-\vec{b}_1-2\vec{b}_2)+y(2\vec{b}_1+2\vec{b}_2)\\
&= (-x+2y)\vec{b}_1+(-2x+2y)\vec{b}_2
\end{aligned}$$

(IV) Find the matrix A_4 of T if the basis B' and B are used for domain and codomain respectively.

First Method

$$\left.\begin{aligned} \left[T(\vec{b}_1)\right]_B &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ \left[T(\vec{b}_2)\right]_B &= \begin{bmatrix} -4 \\ -2 \end{bmatrix} \end{aligned}\right\} \Rightarrow A_4 = \begin{bmatrix} 1 & -4 \\ 1 & -2 \end{bmatrix}$$

$$T\begin{bmatrix} x \\ y \end{bmatrix}_{B'} = A_4\left[\vec{x}\right]_{B'} = \begin{bmatrix} 1 & -4 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}_{B'} = \begin{bmatrix} x-4y \\ x-2y \end{bmatrix}_B$$

Second Method

$$A_1 [\vec{u}]_B = [\vec{v}]_B \implies \underbrace{A_1 P_{B \leftarrow B'}}_{A_4} \underbrace{P_{B' \leftarrow B} [\vec{u}]_B}_{[\vec{u}]_{B'}} = [\vec{v}]_B$$

$$[\vec{u}]_{B'} = [\vec{v}]_B$$

$$A_4 = A_1 P_{B \leftarrow B'} = \begin{bmatrix} -5 & 6 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 1 & -2 \end{bmatrix}$$

$$P_{B' \leftarrow B} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

$$P_{B \leftarrow B'} = - \begin{bmatrix} -1 & -2 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

or

$$A_4 = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 1 & -2 \end{bmatrix}$$

$$A_4 = P_{B \leftarrow B'} A_2$$

$$T(x\vec{b}_1 + y\vec{b}_2) = x T(\vec{b}_1) + y T(\vec{b}_2) =$$

$$= x\vec{b}_1 - 2y\vec{b}_2$$

$$= x(\vec{e}_1 + \vec{e}_2) - 2y(2\vec{e}_1 + \vec{e}_2)$$

$$= (x - 4y)\vec{e}_1 + (x - 2y)\vec{e}_2$$

Third Method

Problem

Let $\mathbb{M}_{22}$ be the space of all 2×2 matrices and let $\mathbb{P}_2$ be the space of all polynomials of at most degree 2. Define $S : \mathbb{P}_2 \to \mathbb{M}_{22}$ as the transformation given by

$$S(a_2x^2 + a_1x + a_0) = \begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix},$$

for all $a_2x^2 + a_1x + a_0 \in \mathbb{P}_2$.

1. *Describe the null space of S. Is S one-to-one?*
2. *Express the range of S as a span of a linearly independent set of 2×2 matrices. Is S onto?*
3. *Is S an isomorphism?*
4. *Extend the basis of the range of S such that the extended basis, $\mathcal{B}_1$, spans $\mathbb{M}_{22}$.*
5. *Consider the ordered basis $\mathcal{B} = \{1, x, x^2\}$ and the ordered basis $\mathcal{B}_1$ for $\mathbb{M}_{22}$ and find the matrix of the transformation S.*
6. *Consider the ordered basis $\mathcal{B} = \{1, x, x^2\}$ and the ordered basis $\mathcal{B}_2 = \left\{ \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$ for $\mathbb{M}_{22}$ and find the matrix of the transformation S.*

1 $\text{Nul } S = \{ p(x) = a_2x^2 + a_1x + a_0 : S(p(x)) = 0_{2\times 2} \}$

$$S(p(x)) = \begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{cases} a_1 + a_2 = 0 \\ a_2 + a_0 = 0 \\ a_1 - a_0 = 0 \\ a_1 + a_0 = 0 \end{cases} \longrightarrow a_2 = a_1 = a_0 = 0$$

which is the trivial Solution.

Hence, null space only contains the zero vector of $\mathbb{P}_2$ which is known as the zero polynomial $p_0(x) = (0)x^2 + 0(x) + 0(1) = 0$ and S is one-to-one.

$$\boxed{\text{Nul } S = \{p_0(x)\}}$$

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2. The output of S is $\begin{bmatrix} a_1 + a_2 & a_2 + a_0 \\ a_1 - a_0 & a_1 + a_0 \end{bmatrix} = a_0 \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}}_{M_0} + a_1 \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}_{M_1} + a_2 \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}}_{M_2}$.

we can check that $\{M_0, M_1, M_2\}$ is a linearly independent set and hence this set is a basis for range of S, since range of S is $\text{span}\{M_0, M_1, M_2\}$. This means $\dim(\text{range of } S) = 3$, whereas $\dim(M_{22}) = 4$. Therefore, S is NOT onto.

3. *Is S an isomorphism?* No. S is not onto.